

# Notes on a Five-Stage Retrieval Pipeline

This page exists to give the extraction stage something safe to read. It is synthetic text, written for this cookbook rather than scanned or copied from any real document, so it can be committed to a public repo with no licensing or privacy question attached to it.

A retrieval pipeline is usually described in five moves: a document is extracted into structured text, the text is chunked into retrievable units, each chunk is embedded into a vector, a query retrieves the closest chunks, and a benchmark scores whether the right ones came back. Each stage inherits the mistakes of the one before it, which is why the extraction stage is worth looking at closely instead of assuming it always works.

Table 1. The five pipeline stages.

| Stage    | Input                    | Output                   |
|----------|--------------------------|--------------------------|
| Extract  | PDF / scanned page       | structured text + tables |
| Chunk    | structured text          | token-aware chunks       |
| Embed    | chunks                   | vectors                  |
| Retrieve | a query                  | ranked chunks            |
| Bench    | questions + gold answers | a score table            |